



Duke University · B.S. Computer Science *Aug 2024 – May 2027 (expected)*

GPA: 4.0 (Dean's List)

Email: zijia.tang@duke.edu | Phone: +1 (858) 340-5757

Research Focus: Embodied AI, efficient VLA, robotic manipulation, AI4Science

Seeking Ph.D. positions starting Fall 2027 in Embodied AI / Robot Learning.

Selected Research & Publications

Robotics / Embodied AI

FLARE: A Failure-Aware Framework for Autonomous Correction and Recovery in Vision-Language-Action Models

G. Zhao, **Z. Tang**, X. Chen, Z. Kuang, Y. Tian, G. Li | *CVPR 2026*

- Proposed a **Retry-and-Reset** recovery framework enabling VLA policies to self-correct after perturbations, including policy failures and external perturbations such as human interventions.
- Addresses the "fragility" problem of VLA models — where minor perturbations can lead to catastrophic failures — enhancing the robustness of VLA models.
- Achieved **84.0% success** on 9 contact-rich RoboMimic tasks, improving prior SOTA by **26.2%**.

RecoverBench: A Benchmark for Error Recovery in Robotic Manipulation

Z. Tang, G. Zhao, X. Chen, et al., G. Li | *Under review, 2026*

- Built a reproducible robotic error-recovery benchmark with **12 error skills**, **136 error subtypes**, and **1,360 validated MuJoCo post-error scenes** across 6 manipulation tasks.
- Designed an automated error-injection mechanism to systematically introduce failures throughout the manipulation, enabling standardized and comprehensive evaluation of robotic error recovery.
- Collected and augmented **1,641 human recovery demonstrations** into **10,598 recovery trajectories**, supporting downstream recovery-policy learning and evaluation.

Latent Bridge: Accelerating Dual-System VLA Models via Latent Feature Prediction

Y. Liu, Y. Li, **Z. Tang**, et al., Y. Chen, H. Li | *arXiv, 2026*

- Developed latent bridge, efficiently predicting visual features to reduce VLM backbone calls while preserving task success rates.
- Achieved up to **75% VLM calls**, resulting in a **1.73 speedup** on LIBERO and RoboCasa.
- Validated across different VLA policies, including GR00T and π -based models, demonstrating a generalizable plug-in approach for accelerating VLM-based robotic control.

Additional Publications

scDrugMap, *Nature Communications (Impact Factor: 15.7)*: foundation-model benchmark for drug-response prediction on **340K+** single cells.

scPerb, *Journal of Advanced Research (Impact Factor: 11.79)*: Built a style-transfer VAE for single-cell perturbation prediction, resulting in **99.5% R^2** correlation with ground truth.

PINet, *ACM-BCB 2024 Rapid-fire Oral*: Built a CNN-Transformer model for Alzheimer's MRI prediction with **96% accuracy**.

Research Experience

Research Assistant, Duke Center for Computational Evolutionary Intelligence — Duke University | *Mar 2026 – Present* | *Supervisors: Prof. Yiran Chen and Prof. Hai Li*

- Worked on efficient inference for dual-system VLA policies, focusing on latent feature prediction, KV-cache modeling, and backbone acceleration.
- Built evaluation pipelines for GR00T and π models across LIBERO and RoboCasa to measure speed–success trade-offs under different VLM-call frequencies.
- Collaborated with senior researchers across model architecture, systems optimization, and embodied AI evaluation.

Research Assistant, HCP Lab — Sun Yat-Sen University | *May 2025 – May 2026* | *Supervisor: Prof. Guanbin Li*

- Worked on failure-aware robotic manipulation, focusing on post-error scene generation, recovery-skill taxonomy, and reproducible recovery evaluation.
- Designed and validated robotic error scenarios across grasp failures, object drops, collisions, wrong-object interactions, and trajectory-regression failures.
- Contributed to recovery data generation and policy evaluation pipelines for VLA-based manipulation systems.

Paid Research Assistant, Yi Zhang Lab — Duke University | *Aug 2024 – May 2025* | *Dept. of Neurosurgery & Dept. of Biostatistics and Bioinformatics*

- Conducted biomedical AI research on single-cell foundation models, cancer knowledge transferring, and medical image analysis.
- Contributed to model development, experiment design, and manuscript preparation for AI-for-science publications.

Research Intern, Song Lab — University of Florida | *May 2023 – Aug 2024* | *Dept. of Health Outcomes & Biomedical Informatics*

- Developed deep learning models for biomedical prediction tasks, including single-cell perturbation response modeling and MRI-based disease prediction.
- Contributed to first author and co-author publications in biomedical AI.

Selected Projects

LeHome Challenge: Deformable Object Manipulation *Team Lead* | *ICRA 2026 Workshop · Competition ongoing*

- Developed cloth-folding policies in the LeHome HI-FI simulation environment, targeting deformable manipulation under low-cost hardware and sim-to-real constraints.

Technical Skills

Robotics / Simulation: MuJoCo, robosuite, robomimic, Isaac Sim, MimicGen, LIBERO, RoboCasa

VLA / Deep Learning: PyTorch, JAX, LoRA fine-tuning, DiT policies, Transformers, KV-cache modeling, VLM inference

Programming: Python, C++, JAVA, Swift, SQL